

# KRISH DILIP PATIL

B.Tech in Computer Science Engineering (3rd Year) • AI/ML Core Focus

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## PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science Engineering student at School of Engineering and Technology (SOET), Pimpri Chinchwad University (PCU) with a 9.3 CGPA. Combines strong foundational computer science principles with practical expertise in Artificial Intelligence, Machine Learning, and full-stack software development. Experienced in designing scalable web applications, real-time computer vision models, and deep learning pipelines.

## EDUCATION

**Pimpri Chinchwad University (PCU) — School of Engineering and Technology (SOET)** **CGPA: 9.3 / 10.0**  
Bachelor of Technology (B.Tech) in Computer Science Engineering (CSE) 2023 – 2027 (Expected)  
Core Coursework: Data Structures & Algorithms, AI/ML, DBMS, Operating Systems, Computer Networks

## TECHNICAL SKILLS

**Programming Languages:** Python, TypeScript, JavaScript, C++, SQL, Java, Bash  
**AI & Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, OpenCV, NLP, LLM API / RAG, Prompt Engineering  
**Full-Stack Development:** React, Node.js, Express, Tailwind CSS, RESTful APIs, WebSockets, HTML5/CSS3  
**Databases & Cloud:** PostgreSQL, MongoDB, Redis, Firebase Firestore, Git, GitHub, Docker, Linux  
**Core Fundamentals:** Data Structures & Algorithms, Object-Oriented Design, System Architecture

## KEY TECHNICAL PROJECTS

- AI Smart Traffic Management & Optimization System** [ Python, PyTorch, YOLOv8, OpenCV, FastAPI, React ]
- Designed real-time deep learning computer vision pipeline analyzing multi-camera CCTV feeds for vehicle classification and density mapping.
  - Developed adaptive signal switching algorithms reducing simulated road intersection congestion by 34%.
  - Implemented asynchronous REST APIs serving sub-85ms model inference and live telemetry dashboards.
- Collaborative Real-Time Code Editor & Cloud IDE** [ React, TypeScript, Node.js, WebSockets, Redis, Docker ]
- Built full-stack browser-based collaborative IDE supporting concurrent multi-user editing with operational transform.
  - Integrated sandboxed remote code execution engine running untrusted programs safely in ephemeral Docker containers.
  - Architected low-latency synchronization with Redis pub/sub handling multiple active rooms and syntax trees.
- LLM Academic Research Paper Synthesis & RAG Search** [ Python, LangChain, ChromaDB, Gemini / OpenAI API, FastAPI ]
- Built retrieval-augmented generation (RAG) system ingesting research papers, chunking text, and storing vector embeddings.
  - Enhanced query retrieval relevance by 40% with hybrid keyword-vector search and citation attribution verification.
  - Created interactive web interface offering real-time streaming answer generation and knowledge graph exploration.
- Distributed In-Memory Key-Value Cache Engine** [ C++, Asynchronous I/O, TCP Sockets, Raft Consensus Protocol ]
- Implemented lightweight, fault-tolerant distributed key-value store modeled on the Raft consensus protocol.
  - Engineered leader election, persistent write-ahead logging (WAL), and snapshot compaction mechanisms.

## LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Technical Lead & Core Team Member — Developer Student Club, SOET PCU** 2024 – Present  
Organized and conducted technical workshops on AI/ML foundations, Git/GitHub, and full-stack web development for 200+ engineering students.
- Hackathon Finalist & Project Lead — Smart India Hackathon & University Level Competitions** 2024  
Led a 4-member engineering team to design and pitch an automated computer-vision quality inspection prototype for smart manufacturing.

## CERTIFICATIONS & HONORS

- Machine Learning Specialization — DeepLearning.AI / Stanford Online
- Full-Stack Web Development Professional Certificate — Meta
- Python for Data Science, AI & Development — IBM
- Postman API Fundamentals Student Expert Certification
- Academic Excellence Award for 9.3 CGPA Standing — Pimpri Chinchwad University SOET